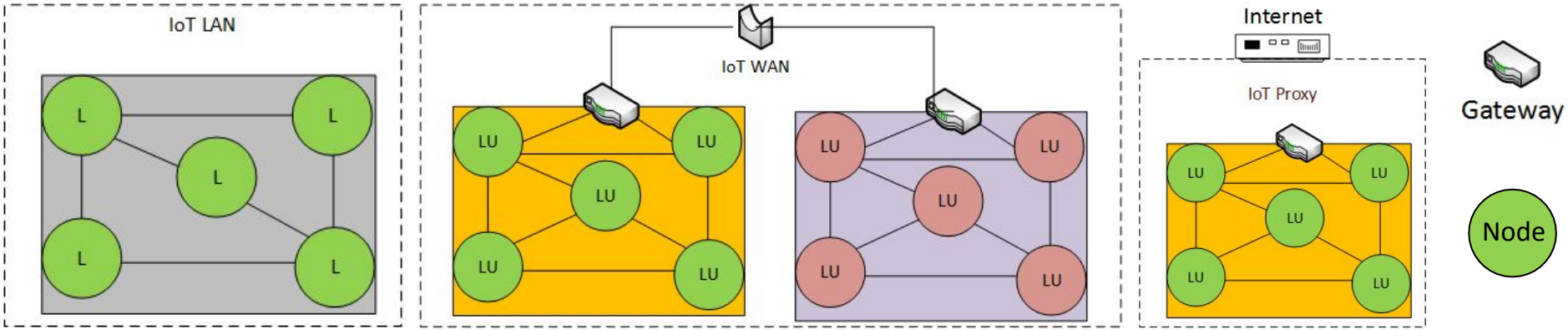
Networking Components of IoT

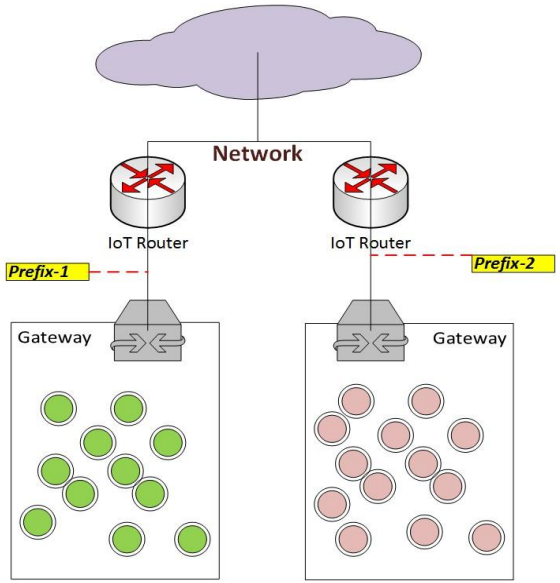
IoT Node	• Connected to other nodes inside a LAN via the IoT LAN, May be sometimes connected to the internet through a WAN directly
IoT Gateway	• A router connecting the IoT LAN to a WAN to the Internet, Can implement several LAN and WAN, Forwards packets between LAN and WAN on the IP layer
IoT LAN	• Enables local connectivity, uses short range communication technologies, May or may not connect to Internet, Building or Organization wide
IoT WAN	• Connects various network segments, Organizationally and geographically wide, operational range between few Kms to hundred Kms, Connects to the internet
IoT Proxy	• Performs active application layer functions between IoT nodes and other entities

IoT Network Configurations



- Some of the IoT network configurations restricted to local areas, analogous to normal LANs, WANs and proxy are shown in the previous figures.
- The nodes represented by green circles have **L:** local link addresses or **LU**: local link addresses which are unique locally.
- Nodes within a gateway's have addresses that are valid within the gateway's domain only.
- The same addresses may be repeated in the domain of another gateway. The gateway have a unique network prefix, which can be used to identify them globally.
- This strategy saves a lot of unnecessary address wastage. Although, the nodes have to communicate to the internet via the gateway.

Gateway Prefix Allotment



- ✔ One of the strategies of address conservation in IoT is to use local addresses which exist uniquely within the domain of the gateway. These are represented by the circles
- ✔ The network connected to the internet has routers with their set of address and ranges.
- ✔ These routers have multiple gateways connected to them which can forward packets from the nodes, to the internet, only via these routers. These routers assign prefixes to gateways under them, so that the gateways can be identified with them.

Addressing strategies in IoT

- In networked devices addressing plays an important role in ensuring the delivery of packets to intended users
- **Data Link Layer Addressing** – concerned with local delivery of frames within a network segment (e.g., MAC addresses)
- Data link layer addressing is divided into three types: Unicast, Multicast, and Broadcast
- **Network Layer Addressing** – concerned with logical addressing to enable end-to-end delivery across networks (e.g., IP addresses).
- **IPv4** primarily emphasizes the reliable delivery of packets between a source and destination
- **IPv6** is fundamentally more address-oriented
- With the rapid growth in the number of devices connected to the Internet, the designers of IPv6 recognized that providing a vast address space was more critical than focusing solely on reliability

IPv4 versus IPv6

	IPv4	IPv6
Developed	IETF 1974	IEF 1998
Length (bits)	32	128
No. of Addresses	2^{32}	2^{128}
Notation	Dotted Decimal	Hexadecimal
Dynamic Allocation of addresses	DHCP	SLAAC/ DHCPv6
IPSec	Optional	Compulsory
Header Size	Variable	Fixed
Header Checksum	Yes	No
Header Options	Yes	No
Broadcast Addresses	Yes	No
Multicast Address	No	Yes

IPv4 Header Format

1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8
VER				HLEN				Type of Service								Total Length															
Identification																Flags				Fragment Offset											
Time to Live								Protocol								Header Checksum															
Source Address (32 bit)																															
Destination Address (32 bit)																															
Options																								Padding							

Field	Description
VER (4 bits)	Represents version of IP
HLEN (4 bits)	Length of IPv4 packet header
ToS (8 bits)	Define priority/quality of service
Total length (16 bits)	Length of IPv4 packet, including header and payload
Identifier (16 bits)	Identify the original packet in case of packet fragmentation
Flags (3 bits)	Control the fragmentation
Fragment offset (13 bits)	Position of fragment in original packet
TTL (8 bits)	Prevents a packet from looping infinitely in the network
Protocol (8 bits)	Identifies the protocol of the packet
Header checksum (16 bits)	Identifying whether a packet is error free or not
Source Address (32 bits)	Indicates source address of the packet
Destination Address (32 bits)	Indicates destination address of the packet
Options & Padding	Optional field

IPv6 Header Format

<table>
<tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr>
<tr><td colspan="4">Ver</td><td colspan="8">Traffic Class</td><td colspan="20">Flow Label</td></tr>
<tr><td colspan="16">Payload Length</td><td colspan="8">Next Header</td><td colspan="8">Hop Limit</td></tr>
<tr><td colspan="32">Source Address (128 bit)</td></tr>
<tr><td colspan="32">Destination Length (128 bit)</td></tr>
</table>

Field	Description
VER (4 bits)	Represents version of IP
Traffic class (8 bits)	Define priority/quality of service
Flow label (20 bits)	designed for streaming media or real time data
Payload length (16 bits)	provides information about packet payload length or amount of data in packet payload
Next header (8bits)	Identifying the type of header
Hop limit (8bits)	Prevents a packet from looping infinitely in the network
Source Address (32 bits)	Indicates source address of the packet
Destination Address (32 bits)	Indicates destination address of the packet

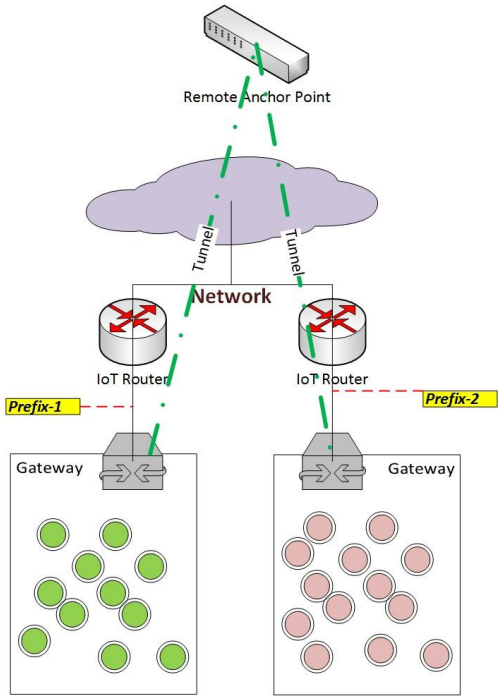
IPv6 Address Format

- IPv6 is a 128 bits long (16 bytes)
- Written in hexadecimal
- Divided into 8 groups of 16 bits (each group = 4 hex digits)
- Groups are separated by colons (:)
- Ex: 2025:9C2E:1234:3F44:4342:AA43:5678:2359

Global prefix Subnet Interface Identifier

- Global Routing Prefix (first 48 bits)→ identifies the organization
- Subnet ID (next 16 bits)→identifies a specific subnet within that organization
- Interface ID (last 64 bits)→ identifies a host/interface on that subnet

Impact of Mobility on Addressing



- In IoT networks, mobility can cause the network prefix to change from 1 to 2. However, the IoT LAN remains unaffected, since the local addresses within the LAN do not change with movement.
- This stability is achieved through the use of Unique Local Addresses (ULA). With ULA, even if the IoT gateway's WAN address changes, the LAN addresses remain the same, ensuring seamless communication inside the local IoT network.

- Gateways are assigned with prefixes and are attached to a remote anchor point. This makes them immune to network prefix changes.
- This stability is achieved using Unique Local Addresses (ULA). The nodes inside a gateway retain their locally unique addresses, so even if the gateway's network prefix changes, the nodes remain unaffected.
- However, in some cases, nodes need to communicate directly with the Internet. This is accomplished through tunneling, where the nodes send packets to a remote anchor point instead of routing through the gateway's router.

- ✔ Despite of providing address stability, ULA cannot communicate directly with the internet or the upper layers, which is solved by implementing an application layer proxy.
- ✔ Application layer proxies may be additionally configured to process data, rather than just passing it.
- ✔ In nodes with no support for computationally intensive tasks, IoT proxy gathers data sent to the link-local multicast address and routes them globally.

- ✔ Presently, the internet is mainly IPv4 based with little or no IPv6 uplink facilities or support.
- ✔ Due to the lack of a universal transition solution to IPv6, lots of un-optimized solutions are being used for IoT deployment.
- ✔ These makeshift solutions mainly address:
  - IPv6 to IPv4 translation
  - IPv6 tunneling over IPv4
  - Application layer proxies (e.g: data relaying)

Multi-homing

- ✔ In cases of small IoT LANs, where allotment of address prefixes is not feasible and possible, a proxy based approach is used to manage multiple IP addresses and map them to link local addresses.
- ✔ In another approach, IoT gateways with their own address prefixes are used for assigning link local addresses to the nodes under it.

Reference: Teemu Savolainen, Jonne Soininen, and Bilhanan Silverajan,*"IPv6 Addressing Strategies for IoT",* IEEE SENSORS JOURNAL, VOL. 13, NO. 10, OCTOBER 2013

- ✔ Providing source addresses, destination addresses and routing information to the multi-homed nodes is the real challenge in multi-homing networks.
- ✔ In case the destination and source addresses originate from the same prefix, routing between gateways can be employed for IoT gateway selection.